SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

SHORT ANSWER TEST

Code of Conduct

I A. Sunitha (192511079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Sunitha.

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

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CO1-AT3 - CLOUD COMPUTING

SHORT ANSWER TEST

1. Define cloud computing and explain any four characteristics that make it different from traditional on premise computing.

2. Trace the evolution of cloud computing from Hardware evolution to internet Software evolution and show how these challenges enable more cloud platforms.

3. Explain role of server virtualization in cloud computing different between virtualization as an enabling technology and cloud computing as a Service model.

4. Discuss the role of data extreme in delivering cloud services.

5. Differentiate IaaS, PaaS and xaaS with one suitable world example for each model.

Answers:

1. **Cloud Computing:**

Cloud computing is the delivery of computing resources such as servers, storage, databases, networking and software over the Internet. users can access. whenever they want or owning.

1. **On-Demand Self-service.**

users can create or access computing resources whenever required without depending on a service provider.

2. **Broad Network Access.**

cloud services can be accessed through the Internet using devices such as laptops, smartphones and tablets.

3. **Resource pooling:**

The provider shares computing resources among multiple users using virtualization.

→Resources are allocated according to the demand.

4. Rapid Scalability:

Resources can be quickly increased or de

based on workload.

For example:-

online shopping website can increase server capacity during festival sale.

2. Trace evolution of cloud computing

cloud computing evoluted through several stages of hardware and software development

Evolution:

1. Mainframe computing:

In the early stages organizations used large maintanance and mainframe computers.

→ Many users are accessed a single powerful computer through the terminals.

2. personal computers:

computing became decentralized with the introduction of PCs.

→ users could perform computing tasks on individual machines.

3. Networking and Internet

The development of computer networks and the Internet allowed computers to communicate and to share resources to different locations.

4. client - server computing

Applications began using seperate client and server systems.

5 Virtualization.

virtualization allowed one physical server to run multiple virtual machines.

6. Internet - Based software

Software started being delivered through Internet insted of being installed locally.

7. Modern cloud computing.

cloud providers combined virtualization, high-speed networks, large data centers. to provide IaaS, PaaS and SaaS.

(3) Role of server virtualization in cloud computing

server virtualization is a technology that divides one physical server into multiple independent virtual machines (VMs) using software called a hypervisor.

Role in cloud computing:

→ Improves utilization of physical hardware.

→ provides isolation between different users

→ Makes it easier to create, move and manage virtual servers.

→ Reduces Infrastructure and maintanance costs.

virtualization	cloud computing
Enabling Technology.	service delivery model
creates mainly focus on efficient resource utilization	Focuses on delivering Services to users.
creates virtual resources from physical resources	provides computing resources through Internet.
Example: 1 physical server → 3 virtual machines → windows, Linux	Example: user → Internet → AWS → virtual Server

→ virtualization provides foundation for efficient cloud Infrastructure.

→ Computing uses technologies such as virtualization to deliver computing services to users.

Role of Data centers

A Data center is facility that will contain more no of servers, storage systems, networking Power Systems.

Roles of Data centers in Delivering cloud services.

1. Computing Resources:

Data centers contains physical server that Provide processing power for cloud applications.

2. Networking:

High speed networking equipment connects services

→ It allows users to access cloud services.

3. Data storage:

→ It will store user files, applications and backups also.

4. Security:

→ usually Data centers use virtualization to create and to manage multiple virtual machines

→ It will provide physical security and encryption monitoring to protect the data.

Virtualization: Data centers use virtualization of Physical servers to create and manage virtual machines.

6. Scalability:-

cloud provides servers and storage when the demand Increases.

7. Realiability:

They will help to keep cloud services available to multiple network connections.

(5) Iaas:- (Infrastructure as a service)

This Iaas provides basic computing Infrastructure such as storage, networking, servers.

Example: AWS.

Paas (platform as a service)

Paas provides development and deployment environment.

→ developers can build applications even though without managing underlying hardware.

Example: Google App Engine.

Saas (Software as service).

Provides complete software applications through Internet.

→ users do not need to install the software.

Example:

Google DOCS.

Xaas.

It will provide everything as a service all the components which are there in this all will provide to service.

Example:

AWS lambda provides computing as a service.

Important points:

→ Iaas provide servers (virtual) and help for networking.

→ paas provides platforms & tools for developing applications.

→ Saas is Ready to use software through Internet.

→ xaas is different from others it resources delivered as services.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow